CyberShield Voice AI

Before You Click, Let CyberShield Check. — voice-first scam protection in Telugu and English, built on Gnani Prisma and Timbre

Great Indian AI Internship Challenge by Gnani.ai · Individual submission by **Maddila Harshitha**, B.Tech CSE (3rd year), Sri Sivani College of Engineering, Srikakulam, Andhra Pradesh

Code: github.com/harshithamaddila818-collab/CyberShield-Voice-AI · Demo video: youtu.be/H3tg2Vn3UFA

PROBLEM

Many people in India receive scam calls, WhatsApp voice notes and SMS in **Telugu** (fake KYC and OTP requests, digital-arrest threats, fake jobs, lottery and electricity-bill scams). Most scam checkers only read English text, and a written warning does not help someone who is more comfortable listening than reading. People need to check a suspicious message in their own language and hear a clear answer.

SOLUTION

CyberShield is **voice-first**: the user records or uploads a suspicious voice message (or pastes text) and gets a risk verdict, the reasons, and a **spoken warning in the language the speaker used**.

1. Listen: Gnani Prisma STT — Telugu / English audio to text, language auto-detected → **2. Decide: rules + Gemini** — Indian scam patterns plus an AI second opinion → **3. Warn: Gnani Timbre TTS** — Warning written for this message, spoken in Telugu or English

Also included: text scan, offline URL structure check, emergency recovery guidance, scan history, downloadable PDF report (Telugu and English), and a per-stage timing strip (Prisma, rules, Gemini, Timbre) so the Gnani steps are visible in the result.

GNANI AI INTEGRATION

- **Prisma STT** (`prisma-v3.0`, `indic-auto` language detection): the entry point of the product. The detected language also selects the language of the spoken warning.
- **Timbre TTS** (`timbre-v2.5`, Telugu and English voices): reads the personalised warning back; a voice scan plays it automatically.
- Not used: Evon (not needed for this use case). Gemini is used for the AI analysis and is disclosed as such.

HOW THE VERDICT IS DECIDED

A rule engine (OTP, KYC, digital arrest, UPI, fake jobs, courier, prize, loan and more, in Telugu, English and Tenglish) gives the first verdict. Gemini adds a second opinion but can only **raise** the score, and alone it is capped at MEDIUM, so an AI mistake cannot create a HIGH alert by itself. A guard keeps genuine OTP notices LOW. If Gemini is unavailable (free-tier quota), the rule result is still shown. Stack: FastAPI, HTML/CSS/JavaScript, SQLite.

VERIFIED RESULTS (ALL SMALL, AUTHOR- OR CONTRIBUTOR-LABELLED; NOT PRODUCTION ACCURACY)

Test	Result
Automated checks (fusion, API, Gemini resilience)	8/8, 18/18 and 9/9 pass
Rules only, fresh messages from 5 other people (30; mostly English)	Precision 100%, recall 53.3%, false alarms 0 of 15 genuine
Rules + Gemini, same 30 messages	Recall 100% (15 of 15), but 2 false alarms in 15 genuine messages (13.3%)
Real voice clips through Prisma (19 clips, one speaker)	Rules on the transcripts: precision 100%, recall 90%, false alarms 0
Unseen scam types, rules only (BLIND4, 24 messages)	Recall 41.7%, which is why Gemini is part of the pipeline

HONEST LIMITATIONS

- Test sets are small (15–30 messages); the true rates are uncertain. The fresh set has almost no Telugu.
- Gemini raised false alarms on legitimate bank alerts.
- Telugu spelling variants in speech output can make a rule miss (one voice-clip scam missed this way).
- For short English clips, Prisma auto-detect sometimes returned Telugu script; word error rate is therefore not reported.
- Telugu PDF letters are readable but not perfectly joined.

NEXT STEPS

- Public hosted demo and tests on mobile devices.
- Larger Telugu test set from many speakers and verified labels.
- Handle Telugu spelling variants and transliterated English before the rules.
- Reduce Gemini false alarms on genuine bank alerts.
- Study with real users.

Data: only made-up or masked messages were used; no real personal information. Assistive tool, not a guarantee that a message is safe or fraudulent.